



Saint Ignatius' College, Riverview Mathematics Assessment Task 2022

Year 12
Mathematics
Task 4 Trial HSC Exam
Date: Thursday 25 th August 2022

General Instructions: • Reading time : 10 mins • Time Allowed: 3 hours • Write using blue or black pen only • NESA approved calculators may be used • Attempt all questions. • A NESA Reference Sheet is provided. • Questions 1 to 10 are all multiple-choice questions worth 1 mark each and are to be answered on the multiple-choice answer sheet provided. • Questions 11 to 35 are each worth 90 marks each and are to be answered on the examination paper. • Each booklet and the multiple-choice answer sheet must have your name and the initials of your class teacher on the front cover. • Marks may not be awarded for missing or carelessly arranged working.	Topics Examined: All Preliminary and HSC Mathematics topics <table style="width: 100%;"> <tr> <td>SECTION 1 Questions 1 -10</td><td style="text-align: right;">10 Marks</td></tr> <tr> <td>SECTION 2 Question 11 -35</td><td style="text-align: right;">90 Marks</td></tr> <tr> <td>Total</td><td style="text-align: right;">100 Marks</td></tr> </table> Teacher: <table style="width: 100%;"> <tr> <td>• Mr N Mushan</td><td style="text-align: right;">NHM</td></tr> <tr> <td>• Mr J Newey</td><td style="text-align: right;">JPN</td></tr> <tr> <td>• Dr M Furtado</td><td style="text-align: right;">MXF</td></tr> <tr> <td>• Ms F Yates</td><td style="text-align: right;">FEY</td></tr> <tr> <td>• Mr S Maher</td><td style="text-align: right;">SJM</td></tr> </table>	SECTION 1 Questions 1 -10	10 Marks	SECTION 2 Question 11 -35	90 Marks	Total	100 Marks	• Mr N Mushan	NHM	• Mr J Newey	JPN	• Dr M Furtado	MXF	• Ms F Yates	FEY	• Mr S Maher	SJM
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SECTION 1 (10 marks)

Attempt Questions 1 – 10

Use the multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9

(A) ☐ (B) ☒ (C) ☐ (D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

correct
↖

1. The domain of $y = \log_e(x + 2)$ is:

- A. $x > 0$
- B. $x > 1$
- C. $x > -3$
- D. $x > -2$

2. Compared with the graph of $y = \cos x$, the graph of $y = \cos \frac{x}{2}$ has:

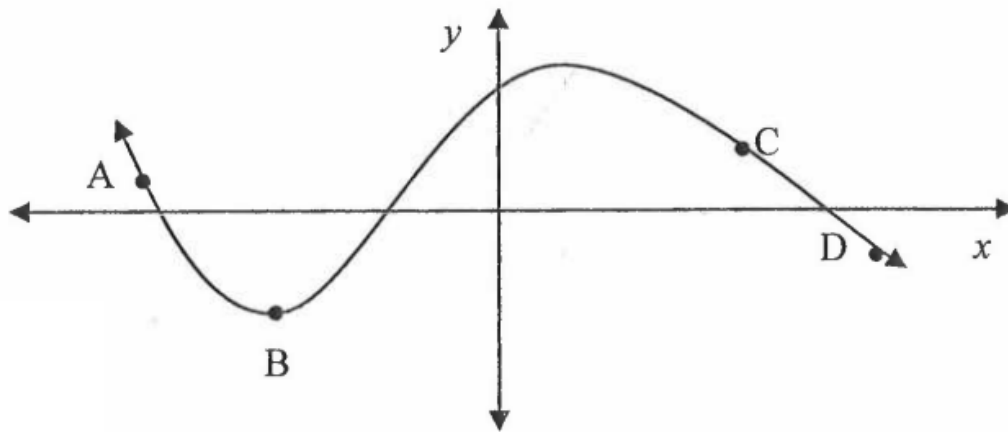
- A. half the amplitude but the same period
- B. the same amplitude and half the same period
- C. double the amplitude and the same period
- D. the same amplitude and double the period

3. A set of data is normally distributed with a mean of 8.6 and a standard deviation of 0.7. The percentage of scores that lie between 8.6 and 9.3 is

- A. 17.5%
- B. 34%
- C. 68%
- D. 95%

4. Which point on the following diagram relates to the following description.

$$y > 0, \frac{dy}{dx} < 0, \frac{d^2y}{dx^2} < 0$$



- A. A
- B. B
- C. C
- D. D

5. $\int_1^a 4-2x \, dx = -3$ where $a > 0$. The value of a is

A. 1

B. 4

C. 6

D. 8

6. $\int_0^1 e^{3x} \, dx$

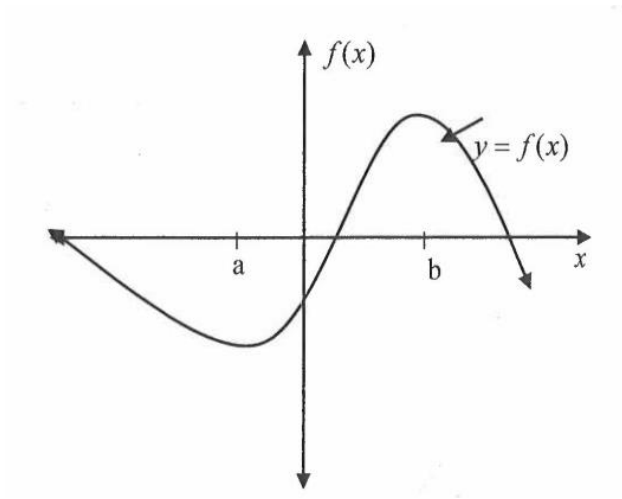
A. $1-e^3$

B. e^3-1

C. $\frac{e^3-1}{3}$

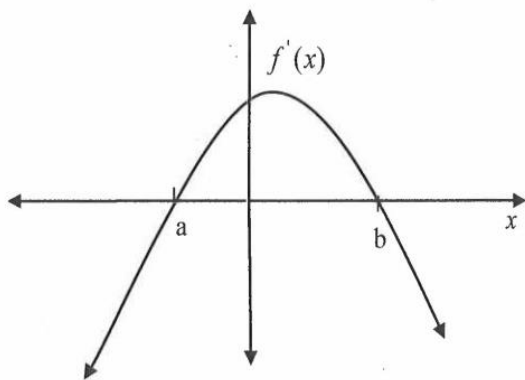
D. $\frac{1-e^3}{3}$

7. The diagram shows the graph of $y = f(x)$

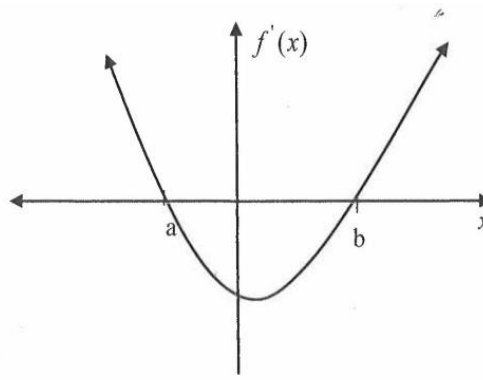


Which of the following is the graphs shows $y = f'(x)$?

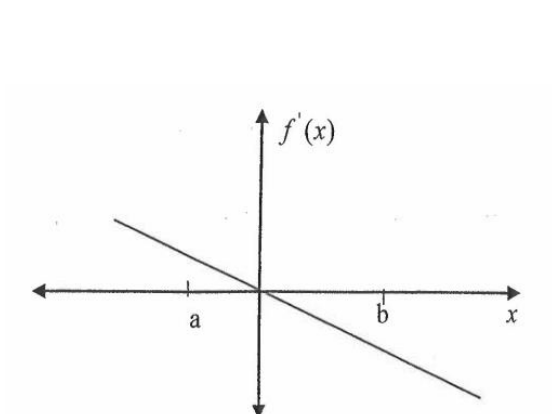
A.



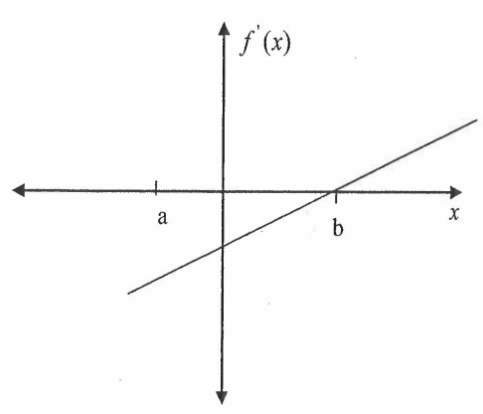
B.



C.



D.



8. The solutions to $e^{6x} - 5e^{3x} + 6 = 0$ are $x =$

A. 2, 3

B. $\log_e 2, \log_e 3$

C. e^{2x}, e^{3x}

D. $\frac{1}{3}\log_e 3, \frac{1}{3}\log_e 2$

9. If $y = \log_a x$, then $\frac{dy}{dx} =$

A. $\frac{1}{x}$

B. $\frac{1}{x \log_e a}$

C. $\frac{1}{a}$

D. $\frac{1}{\log_e x}$

10. Given the curve $y = f(x)$, $\frac{d^2y}{dx^2} = (x+4)^2(x-2)$, which of the following statements are correct

A. $x = -4$ and $x = 2$ are both the x co-ordinates of points of inflection

B. $x = -4$ is the only x co-ordinate of a point of inflection

C. $x = 2$ is the only x co-ordinate of a point of inflection

D. $x = -4$ and $x = 2$ are both not the x co-ordinates of points of inflection

End of Section 1

SECTION 2**Total Marks – 90****Attempt Questions 11-31****[Marks for each part are indicated on the page]****Allow about 2 hours and 45 minutes for this section****Question 11****Marks**

A rainwater tank which is full, is drained so that at time ' t ' minutes the volume of water V in litres is given by $V = 500 \left(1 - \frac{t}{60}\right)^2$ for $0 \leq t \leq 60$.

(a) How much water was initially in the tank?

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(b) After how many minutes was the tank half full? (correct to 2 decimal places)

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(c) At what rate was the water draining when the time is 58 minutes.

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Question 12

Find the sum of the series $8+14+20+26+ \dots +152$.

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Question 13

Evaluate $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$.

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Question 14

The ratio of boys to girls at Rivendale High School is 3:5. The ratio of boys to girls in South Park High school is 2:3. A student is chosen at random from Rivendale High School and then another student is chosen from South Park High School.

Find the probability that:

(a) two boys are chosen.

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(b) at least one girl is chosen.

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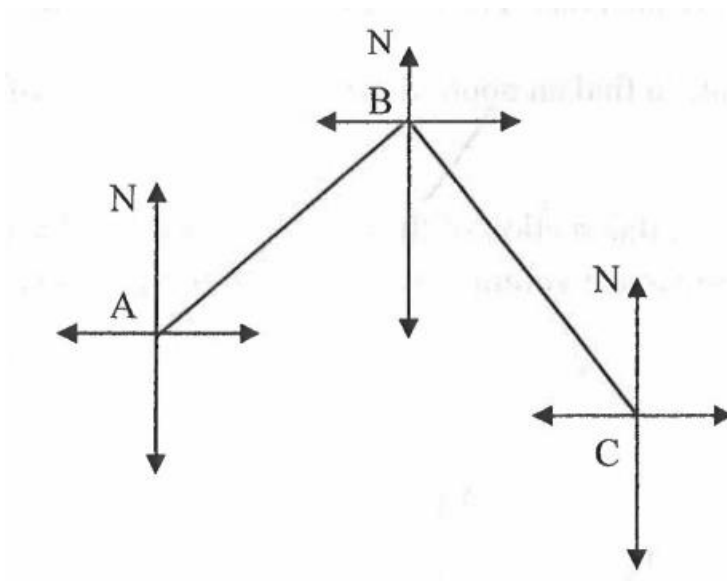
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Question 15

A ship sails 150 km from Appleville (A) to Brooktown (B) on a bearing of 050° T. It then sails on a bearing of 130° T to Cook (C) which is 300 km from Brooktown.



Label the diagram above showing all the essential features

1

(a) Find the distance from Appleville (A) to Cook (C)
(Give your answer correct to two decimal places)

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(b) What is the bearing of Appleville (A) from Cook (C)?
(Give your answer to the nearest minute)

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Question 16

For the curve $y = \frac{x^3}{3} - 3x^2 + 8x + 5$

(a) Find the coordinates of the stationary points and distinguish their nature.

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(b) Find the point of inflection.

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(c) What is the maximum value of $y = \frac{x^3}{3} - 3x^2 + 8x + 5$ for the domain $x \in R [0, 6]$?

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Question 17

(a) Evaluate $\int_2^3 \frac{x}{x^2-1} dx$. (Write your answer as an exact value)

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(b) Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos x}{\sin x} dx$. (Write your answer as an exact value)

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Question 18

Differentiate $\frac{x}{\tan 2x}$.

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Question 19

Expand and simplify $(\operatorname{cosec} A + 1)(\operatorname{cosec} A - 1)$.

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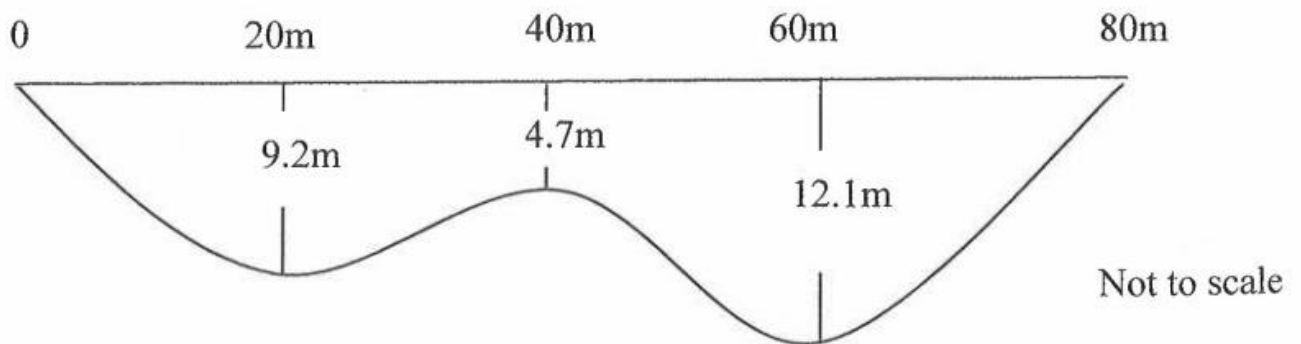
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Question 20

The following diagram represents a cross-section through a river.
The depth of the river is marked every 20 metres.



Use the trapezoidal rule with 5 function values to estimate the area of the cross-section.

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Question 21

- (a) The temperature in the kitchen of Nameeta's new home is controlled by a thermostat.

The temperature can be modelled by the equation: $T = 21.7 + \sin\left(\frac{t}{2}\right)$ where

T is the temperature in degrees Celcius and t is the time in minutes since Nameeta entered the kitchen.

How long will Nameeta have to wait before the temperature gets to 22.5°C

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- (b) The volume $V \text{ cm}^3$ of a balloon is increasing such that the volume at any time t seconds is given by $V = \frac{\pi t^3}{3} - \frac{\pi t^2}{6} + \frac{1}{2}$.

Find the rate at which the volume is increasing when $t = 2$

2

(write your answer in exact form)

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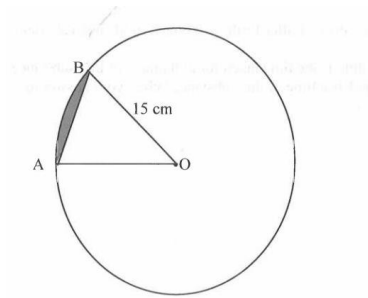
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Question 22

The diagram is a circle centre O and radius 15 cm. The triangle AOB is an equilateral triangle.

AB is a chord of the circle.

(a) State in radians the size of angle AOB.

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(b) Calculate the area of the shaded region. (Write your answer as an exact value)

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Question 23

A uniform continuous random variable with probability density function is shown below.

$$f(x) = \begin{cases} ax^2, & 0 \leq x \leq 6 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of a .

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(b) Find $P(3 \leq X \leq 5)$

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Question 24

The circle $x^2 + 6x + y^2 - 4y - 3 = 0$ is reflected in the x -axis.

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Sketch **on the next page** the **reflected** circle, showing the coordinates of the centre and the radius.

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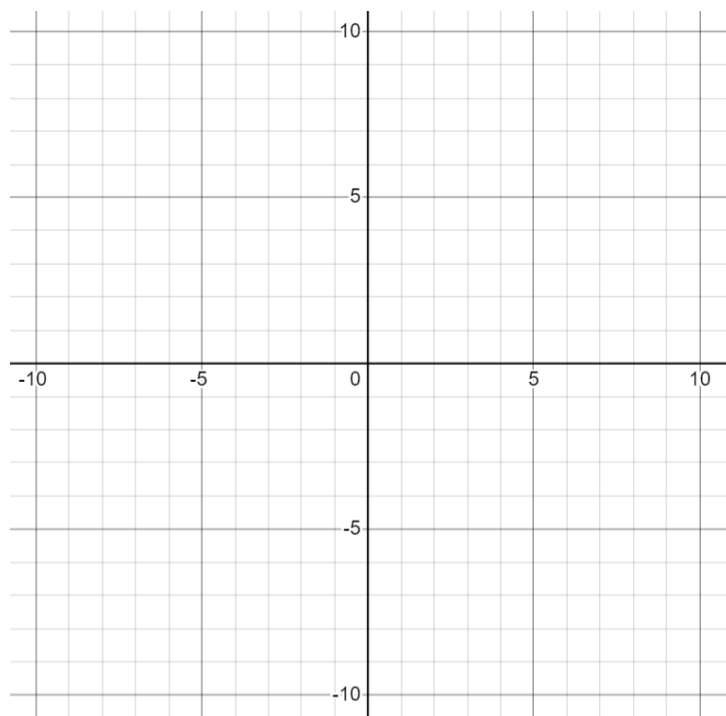
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Question 25

Jason deposits an amount of \$100 000 into an account which pays compound interest of 2% per annum, added to the account at the end of each year.

Immediately after the interest is added, Jason makes a withdrawal for expenses for the coming year. The first withdrawal is \$M. Each subsequent withdrawal is 10% greater than the previous one.

Let A_n be the amount of money in the account after the n th withdrawal.

(a) Show that $A_2 = 100\,000(1.02)^2 - M(1.02 + 1.10)$

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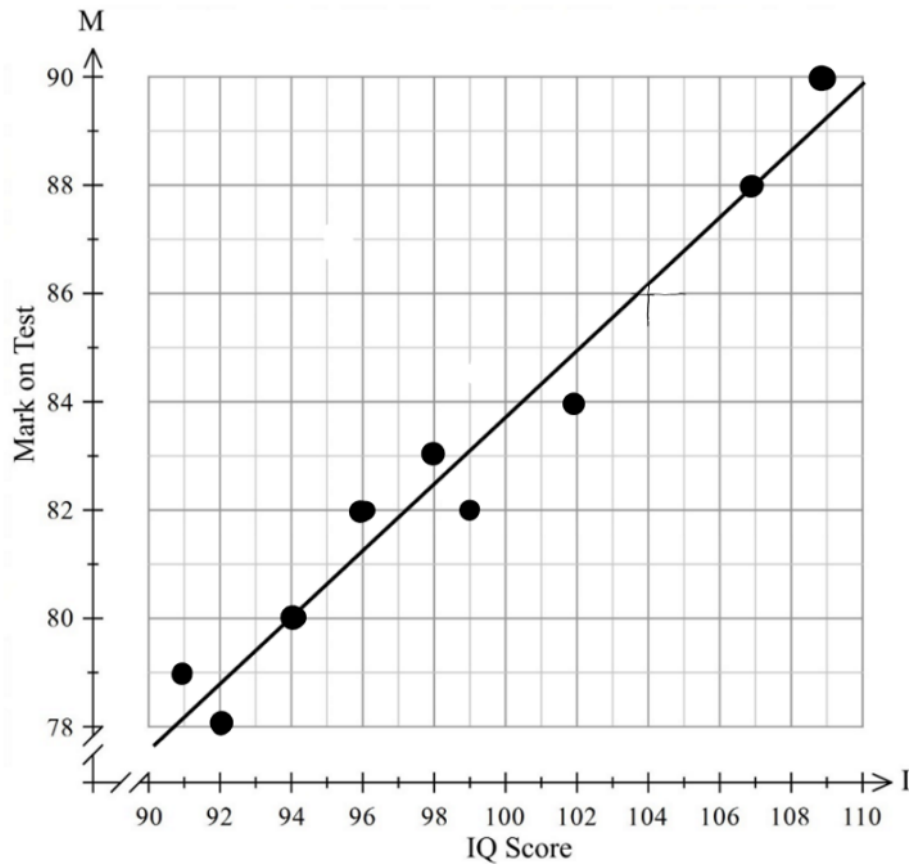
(b) Show that $A_3 = 100\,000(1.02)^3 - M[(1.02)^2 + (1.02)(1.10) + (1.10)^2]$

1

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Question 26

The bivariate data in the scatterplot below compare the recorded IQ scores of students with their mark on a class test out of 100. A line of best fit was also drawn.



(Note: Both axes are truncated)

(a) Calculate the gradient of the line of best fit

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(b) Describe the correlation between the IQ score and the mark on the test in terms of **direction** and **strength**.

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Question 27

A continuous probability distribution is given by $f(x) = \frac{3x^2}{279}$ defined in the domain $[4, 7]$

(a) Find the cumulative distribution function.

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(b) Find the median of the continuous probability distribution.

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Question 28

Evaluate $\int_1^3 x^2(x^3 + 8)^2 dx$

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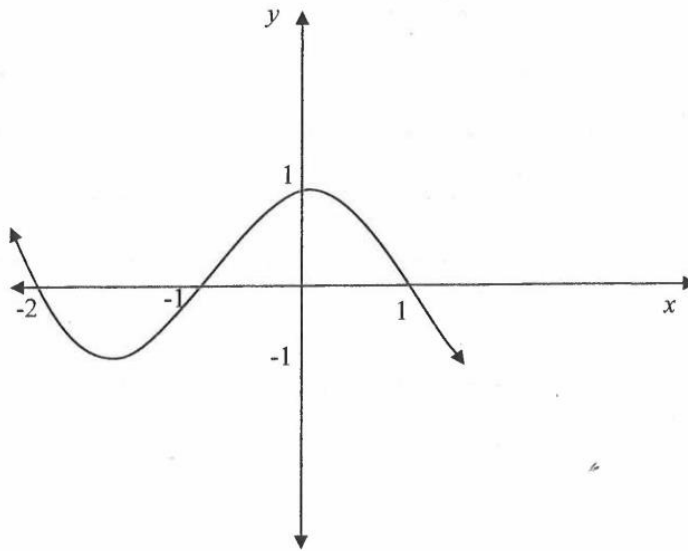
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Question 29

(a) The graph of $y = f(x)$ is shown.



On the graph above draw the curve, $y = f(x-1)+1$, showing all important features.

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Question 30

A class of students sat a test worth 30 marks. The marks were normally distributed.

The mean was 18 and the standard deviation was 1.6.

(a) Calculate Julie's z-score if her mark was 22.

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(b) If Dan's z-score was 1.25, calculate his actual mark.

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Question 31

A psychologist proposes that the ability of a child to memorise during the first five years can be modelled by the function $f(x) = 1 + x \log_e x$ $0 < x \leq 5$

(i.e the ability to memorise at age x years is $f(x)$)

(a) During which month is the ability at a minimum in the first five years

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(b) When is it a maximum during this period. Give a reason for your answer.

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Question 32

Evaluate $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y$, given that $y = xe^{-2x}$.

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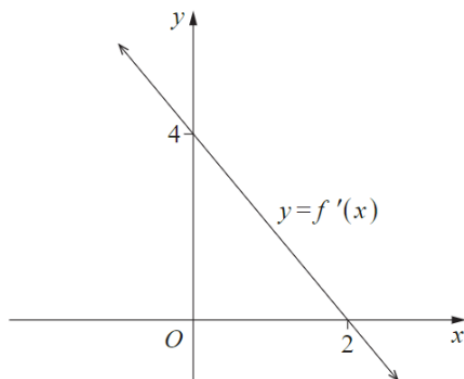
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Question 33

The graph of the derivative of a function $y = f'(x)$ is shown below.

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The curve $y = f(x)$ has a maximum value of 12.

What is the equation of $y = f(x)$?

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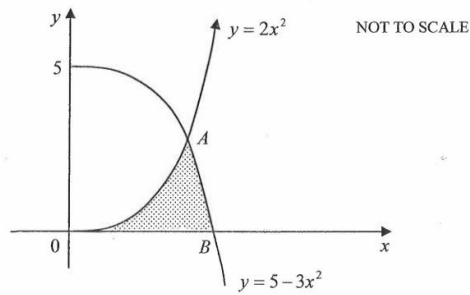
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Question 34



The shaded region OAB is bounded by the parabolas $y=2x^2$, $y=5-3x^2$ and the x -axis.

Point A is the intersection of the two parabolas and point B is the x -intercept of the parabola $y=5-3x^2$

- (a) Show that the x - coordinates of A and B are 1 and $\frac{\sqrt{5}}{\sqrt{3}}$ respectively.

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- (b) Find the exact area of the shaded region OAB.

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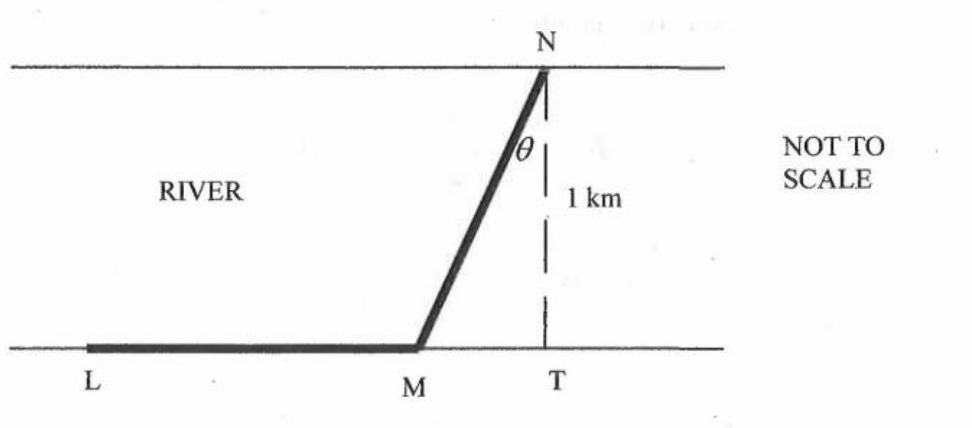
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Question 35

It is desired to construct a cable link between two points L and N, which are situated on opposite banks of a river of width 1 km. L lies 3 km upstream from N. It costs 3 times as much to lay a length of cable underwater as it does to lay the same cable overland.

The following diagram is a sketch of cables where θ is the angle where NM makes with the direct route across the river.



Note: $MN = \sec \theta$ and $MT = \tan \theta$

(a) If segment LM costs c dollars per km, prove the total cost (T) of laying the cable is given by $T = 3c - c \tan \theta + 3c \sec \theta$

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(b) At what angle should the cable cross the river in order to minimise the total cost of laying it. 3

[illegible]

End of Exam



SUGGESTED
SOLUTIONS

Saint Ignatius' College, Riverview

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2022

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Mathematics
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Topics Examined:

All Preliminary and HSC Mathematics topics

SECTION 1

Questions 1 -10

10 Marks

SECTION 2

Question 11 -35

90 Marks

Total

100 Marks

Teacher:

- | | | |
|----------------|------------------|-----|
| • Mr N Mushan | Q32 - Q35 | NHM |
| • Mr J Newey | Q16 - Q21 | JPN |
| • Dr M Furtado | Q26 - Q31 | MXF |
| • Ms F Yates | Q22 - Q25 | FEY |
| • Mr S Maher | Q11 - Q15 | SJM |

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SECTION 1 (10 marks)

Attempt Questions 1 – 10

Use the multiple-choice answer sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

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(A) ☐ (B) ☒ (C) ☐ (D) ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

(A) ☒ (B) ☒ (C) ☐ (D) ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows.

(A) ☒ (B) ☒ (C) ☐ (D) ☐ *correct*

1. The domain of $y = \log_e(x+2)$ is:

A. $x > 0$

B. $x > 1$

C. $x > -3$

☒ D. $x > -2$

$$\textcircled{1}: x+2 > 0$$

$$x > -2$$

2. Compared with the graph of $y = \cos x$, the graph of $y = \cos \frac{x}{2}$ has:

A. half the amplitude but the same period

B. the same amplitude and half the same period

C. double the amplitude and the same period

☒ D. the same amplitude and double the period

$$y = \cos x$$

$$\text{period} = 2\pi$$

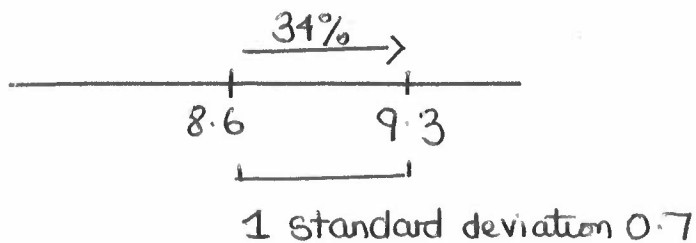
$$y = \cos \frac{x}{2}$$

$$\text{period} = \frac{2\pi}{\frac{1}{2}}$$

$$= 4\pi$$

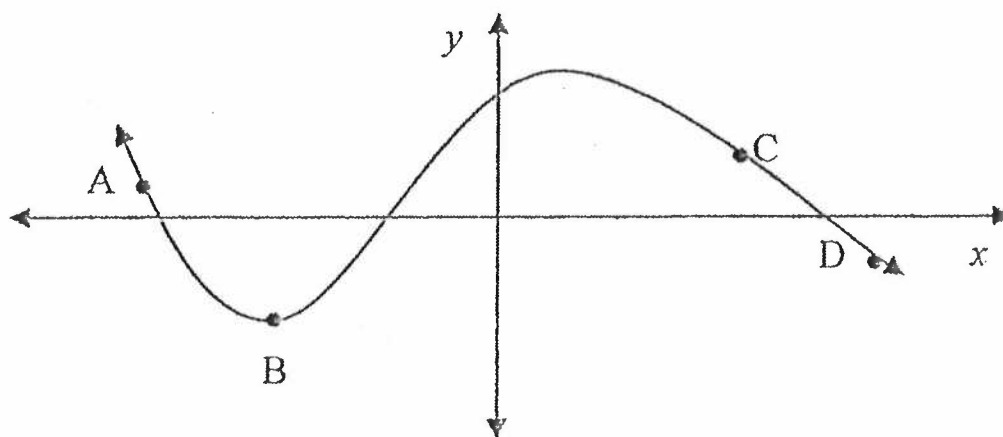
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- A. 17.5%
☒ B. 34%
 C. 68%
 D. 95%



4. Which point on the following diagram relates to the following description.

$$y > 0, \frac{dy}{dx} < 0, \frac{d^2y}{dx^2} < 0$$



- A. A
 B. B
☒ C. C
 D. D

$$y > 0 \quad A \text{ or } C$$

$$\frac{dy}{dx} < 0 \quad A \text{ or } C \quad (\text{curve is decreasing})$$

$$\frac{d^2y}{dx^2} < 0 \quad \textcircled{C} \quad (\text{concave down})$$

5. $\int_1^a 4-2x \, dx = -3$ where $a > 0$. The value of a is

A. 1

☒ B. 4

C. 6

D. 8

$$\left[4x - x^2 \right]_1^a = -3$$

$$(4a - a^2) - (4 - 1) = -3$$

$$a^2 - 4a = 0$$

$$a = 0 \text{ or } 4$$

6. $\int_0^1 e^{3x} \, dx$

$$\frac{1}{3} \left[e^{3x} \right]_0^1 = \frac{1}{3} (e^3 - 1)$$

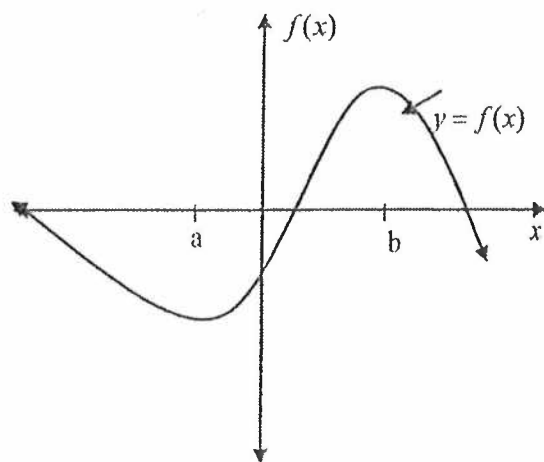
A. $1 - e^3$

B. $e^3 - 1$

☒ C. $\frac{e^3 - 1}{3}$

D. $\frac{1 - e^3}{3}$

7. The diagram shows the graph of $y = f(x)$



original function: 'cubic'
 derivative function: 'quadratic'
 A or B

S.P at $x=a$ and b
 derivative function 0 (at a and b)
 A or B

Nature of SP

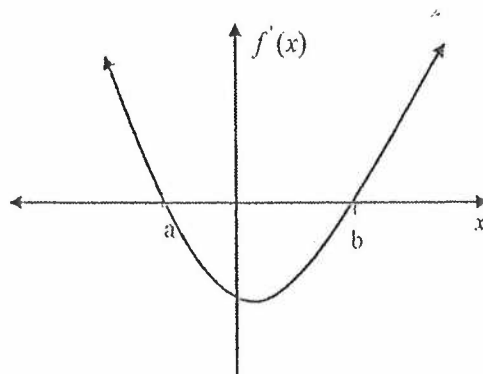
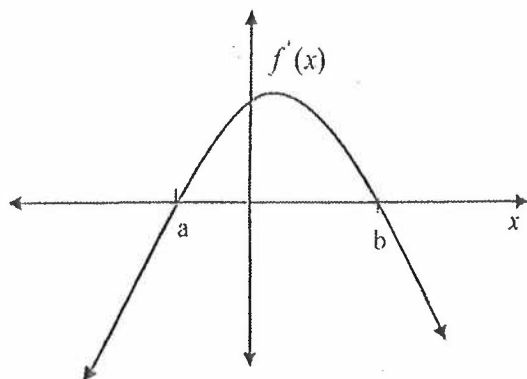
$x=a$ (min T.P) $\cup$
 $x=b$ (max T.P) $\cap$

Which of the following is the graphs shows $y = f'(x)$?

A.

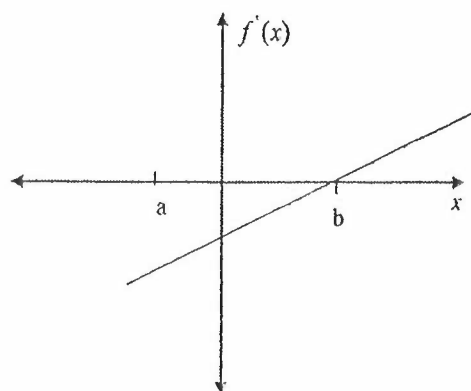
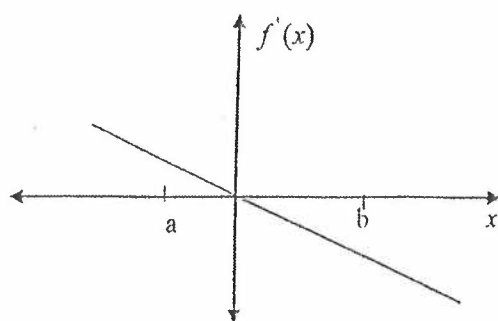
B.

∴ A



C.

D.



8. The solutions to $e^{6x} - 5e^{3x} + 6 = 0$ are $x =$

A. 2, 3

B. $\log_e 2, \log_e 3$

C. e^{2x}, e^{3x}

☒ D. $\frac{1}{3}\log_e 3, \frac{1}{3}\log_e 2$

$$u^2 - 5u + 6 = 0$$

$$(u-2)(u-3) = 0$$

$$u = 2$$

$$u = 3$$

$$e^{3x} = 2$$

$$e^{3x} = 3$$

$$3x = \ln 2 \text{ or } \ln 3$$

9. If $y = \log_a x$, then $\frac{dy}{dx} =$

A. $\frac{1}{x}$

☒ B. $\frac{1}{x \log_e a}$

C. $\frac{1}{a}$

D. $\frac{1}{\log_e x}$

$$y = \frac{\ln x}{\ln a}$$

$$\frac{dy}{dx} = \frac{1}{\ln a} \cdot \frac{1}{x}$$

10. Given the curve $y = f(x)$, $\frac{d^2y}{dx^2} = (x+4)^2(x-2)$, which of the following statements are correct

A. $x = -4$ and $x = 2$ are both the x co-ordinates of a point of inflection

B. $x = -4$ is the only x co-ordinate of a point of inflection

☒ C. $x = 2$ is the only x co-ordinate of a point of inflection

D. $x = -4$ and $x = 2$ are both not the x co-ordinates of a point of inflection

$$x = -5, y'' = -7$$

$$x = 0, y'' = -32$$

$$x = 3, y'' = 49$$

End of Section 1

SECTION 2

Total Marks – 90

Attempt Questions 11-31

[Marks for each part are indicated on the page]

Allow about 2 hours and 45 minutes for this section

Question 11

Marks

A rainwater tank which is full is drained so that a time ' t ' minutes the volume of water V in litres is given by $V = 500 \left(1 - \frac{t}{60}\right)^2$ for $0 \leq t \leq 60$.

(a) How much water was initially in the tank?

1

$$V = 500 \text{ L}$$

1 mk

(b) After how many minutes was the tank half full?

2

$$\left(1 - \frac{t}{60}\right)^2 = \frac{1}{2} \quad \text{link for progress}$$

$$t = 17.57 \text{ (18 min)} \quad \text{or} \quad t = 102.4 \text{ (reject)}$$

link for correct solution

$$0 \leq t \leq 60$$

(c) At what rate was the water draining when the time is 58 minutes.

2

$$\frac{dV}{dt} = 1000 \left(1 - \frac{t}{60}\right) \times -\frac{1}{60} \quad \text{link for derivative}$$

$$\text{when } t = 58$$

$$\frac{dV}{dt} = -\frac{5}{9} \quad \text{(draining at } \frac{5}{9} \text{ L/min)}$$

link for answer

Question 12

Find the sum of the series $8+14+20+26+\dots+152$.

$$a + (n-1)d = 152$$

$$8 + (n-1)6 = 152$$

$$n = 25$$

link for n

$$\text{A.P. } a = 8 \quad d = 6$$

$$S_{25} = \frac{25}{2} [8 + 152]$$

$$= 2000 \quad \text{link for sum}$$

Question 13

Evaluate $\int_0^{\frac{\pi}{8}} \sec^2 2x \, dx$.

2

$$\frac{1}{2} \left[\tan 2x \right]_0^{\frac{\pi}{8}} \quad \checkmark \quad \text{link for integration}$$

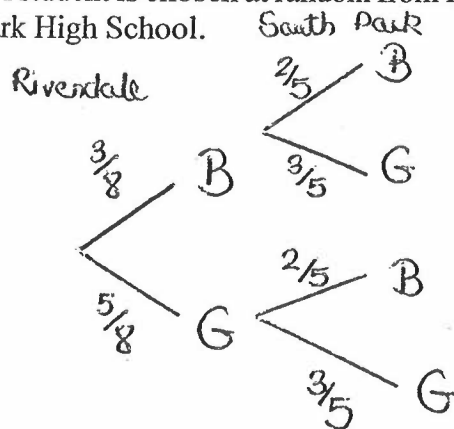
$$= \frac{1}{2} \left[\tan \frac{\pi}{4} - \tan 0 \right]$$

$$= \frac{1}{2} [1 - 0]$$

$$= \frac{1}{2} \quad \checkmark \quad \text{link for substitution and answer}$$

Question 14

The ratio of boys to girls at Rivendale High School is 3:5. The ratio of boys to girls in South Park High school is 2:3. A student is chosen at random from Rivendale High School and then another student is chosen from South Park High School.



link for method to determine $\frac{3}{8}$ and $\frac{2}{5}$

Find the probability that:

(a) two boys are chosen.

2

$$P(BB) = \frac{3}{8} \times \frac{2}{5}$$

$$= \frac{3}{20} \quad \checkmark \quad \text{link for solution}$$

(b) at least one girl is chosen.

1

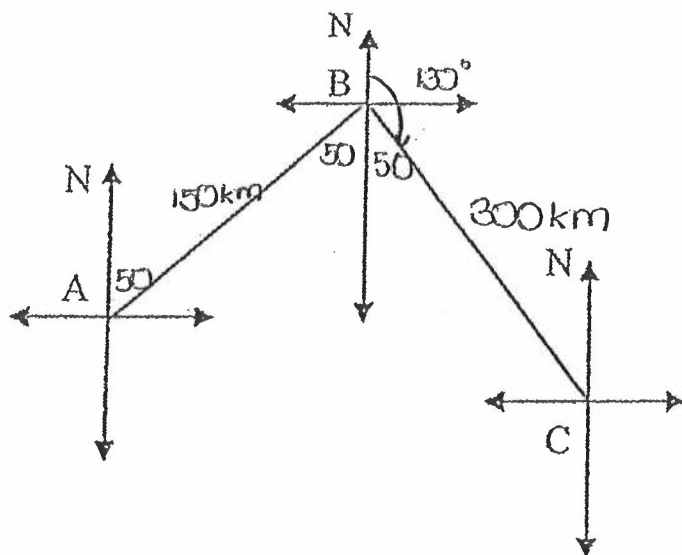
$$1 - P(BB)$$

$$1 - \frac{3}{20}$$

$$= \frac{17}{20} \quad \checkmark \quad \text{link for using complement to calculate answer}$$

15

A ship sails 150 km from Appleville (A) to Brooktown (B) on a bearing of 050° T. It then sails on a bearing of 130° T to Cook (C) which is 300 km from Brooktown.



Label the diagram above showing all the essential features

1

(a) Find the distance from Appleville (A) to Cook (C)
(Give your answer correct to two decimal places)

2

$$\angle ABC = 50 + 50 = 100^\circ$$

$$AC^2 = 150^2 + 300^2 - 2(300)(150)\cos 100^\circ \quad \text{link for correct use of cosine rule}$$

$$b = \frac{375.95}{357.95} \quad (2 \text{ d.p.}) \quad \text{link for correct solution}$$

(b) What is the bearing of Appleville (A) from Cook (C)?
(Give your answer to the nearest minute)

2

$$\frac{\sin \angle BCA}{50} = \frac{\sin 100}{357.95}$$

$$\sin \angle BCA = \frac{50 \sin 100}{357.95}$$

$$\angle BCA = 24^\circ 22' \quad \text{link for } \angle BCA$$

$$\begin{aligned} \text{Bearing} &= 360 - (50 + 24^\circ 22') \\ &= 285^\circ 38' \quad \text{link for bearing} \end{aligned}$$

Question 16

For the curve $y = \frac{x^3}{3} - 3x^2 + 8x + 5$

(a) Find the coordinates of the stationary points and distinguish their nature.

3

$$y' = x^2 - 6x + 8$$

for stationary points let $y' = 0$

$$x^2 - 6x + 8 = 0$$

$$(x-2)(x-4) = 0$$

$$x = 2 \text{ or } 4$$

✓ finding x co-ords of stat points.

x	0	2	3
y'	8	0	-1

x	3	4	5
y'	-1	0	3

✓ co-ordinates of both stationary points



$\therefore (2, \frac{35}{3})$ is a
maximum turning point

$(4, \frac{31}{3})$ is a
minimum turning point

✓ determining their nature.

(b) Find the point of inflection.

2

$$y'' = 2x - 6$$

for possible P.O.I let $y'' = 0$

$$2x - 6 = 0$$

$$2x = 6$$

$$x = 3$$

test

x	0	3	4
y''	-6	0	2

✓ showing an appropriate test.

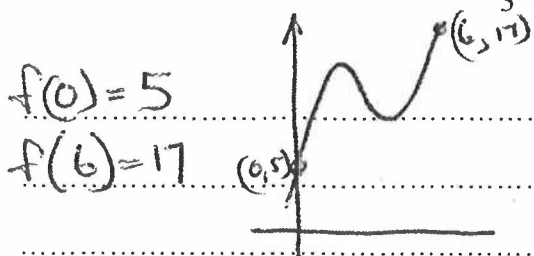
since y'' changes sign
 $\therefore$ concavity changes

$\therefore (3, 11)$ is a
point of inflection

✓ point of inflection

(c) What is the maximum value of $y = \frac{x^3}{3} - 3x^2 + 8x + 5$ for the domain $x \in R [0, 6]$?

2



$$f(0) = 5$$

$$f(6) = 17$$

The maximum value of the
function in the given domain is 17 ✓✓

(only 1 mark for (6, 17))

Question 17

(a) Evaluate $\int_2^3 \frac{x}{x^2-1} dx$. (Write your answer as an exact value)

2

$$\begin{aligned}
 &= \frac{1}{2} \int_2^3 \frac{2x}{x^2-1} dx \\
 &= \frac{1}{2} \left[\ln |x^2-1| \right]_2^3 \quad \checkmark \text{ for integrating correctly} \\
 &= \frac{1}{2} (\ln 8 - \ln 3) \quad \checkmark \\
 &= \frac{1}{2} \ln \frac{8}{3} \quad (\text{or equivalent eg } \ln \sqrt{\frac{8}{3}})
 \end{aligned}$$

(b) Evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \frac{\cos x}{\sin x} dx$. (Write your answer as an exact value)

3

$$\begin{aligned}
 &= \left[\ln |\sin x| \right]_{\frac{\pi}{6}}^{\frac{\pi}{2}} \quad \checkmark \text{ for integrating correctly} \\
 &= \ln \left| \sin \frac{\pi}{2} \right| - \ln \left| \sin \frac{\pi}{6} \right| \\
 &= \ln 1 - \ln \frac{1}{2} \quad \checkmark \\
 &= -\ln \frac{1}{2} \quad \checkmark = \ln \left(\frac{1}{2} \right)^{-1} = \ln 2
 \end{aligned}$$

Question 18

Differentiate $\frac{x}{\tan 2x}$.

2

$$\begin{aligned}
 &\text{let } u = x \quad v = \tan 2x \\
 &u' = 1 \quad v' = 2 \sec^2 2x \quad \checkmark \text{ differentiating correctly} \\
 &y' = \frac{\tan 2x - 2x \sec^2 2x}{\tan^2 2x} \quad \checkmark \text{ correctly using quotient rule.}
 \end{aligned}$$

Question 19

Expand and simplify $(\operatorname{cosec} A + 1)(\operatorname{cosec} A - 1)$.

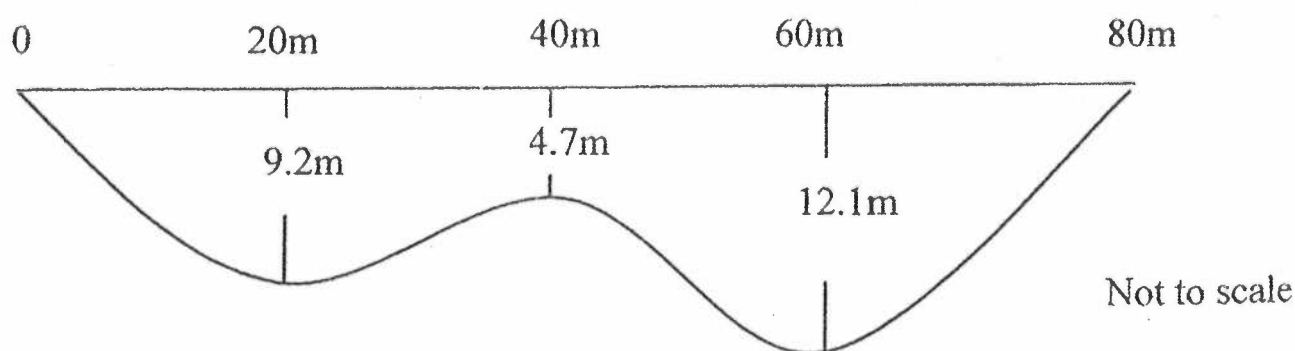
2

$$= \operatorname{cosec}^2 A - 1 \quad \checkmark \quad \text{expanding correctly}$$

$$= \cot^2 A \quad \checkmark \quad \text{simplifying}$$

Question 20

The following diagram represents a cross-section through a river.
The depth of the river is marked every 20 metres.



Use the trapezoidal rule with 5 function values to estimate the area of the cross-section.

2

$$A \approx \frac{h}{2} [f(a) + f(b) + 2(f(x_1) + f(x_2) + f(x_3))]$$

$$\approx \frac{20}{2} [0 + 0 + 2(9.2 + 4.7 + 12.1)] \quad \checkmark$$

$$\approx \boxed{520 \text{ m}^2} \quad \checkmark$$

note h = width of each strip

$$A = \frac{b-a}{2n} [f(a) + f(b) + 2(f(x_1) + f(x_2) + f(x_3))]$$

n is the number of sub intervals (ie no. of strips)

Question 21

- (a) The temperature in the kitchen of Nameeta's new home is controlled by a thermostat.

The temperature can be modelled by the equation: $T = 21.7 + \sin\left(\frac{t}{2}\right)$ where

T is the temperature in degrees Celcius and t is the time in minutes since Nameeta entered the kitchen.

How long will Nameeta have to wait before the temperature gets to 22.5°C

2

$$22.5 = 21.7 + \sin\left(\frac{t}{2}\right)$$

$$0.8 = \sin \frac{t}{2}$$

$$\frac{t}{2} = 0.92729\dots$$

$$t = 1.8545\dots \text{ minutes}$$

many forgot to
use RADIANS !!

(1 min 51 seconds)

- (b) The volume $V \text{ cm}^3$ of a balloon is increasing such that the volume at any time t seconds is given by $V = \frac{\pi t^3}{3} - \frac{\pi t^2}{6} + \frac{1}{2}$.

Find the rate at which the volume is increasing when $t=2$

2

(write your answer in exact form)

$$\frac{dV}{dt} = \pi t^2 - \frac{\pi t}{3}$$

finding $\frac{dV}{dt}$

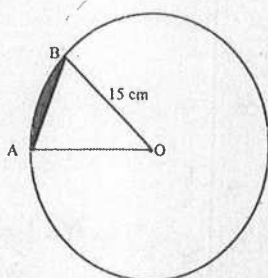
when $t=2$

$$\frac{dV}{dt} = \pi \times 2^2 - \frac{\pi \times 2}{3}$$

$$= 4\pi - \frac{2\pi}{3}$$

$$= \boxed{\frac{10\pi}{3}} \text{ cm}^3 \text{ s}^{-1}$$

Question 22



The diagram is a circle centre O and radius 15 cm. The triangle AOB is an equilateral triangle.

AB is a chord of the circle.

* Some failed to see

(a) State in radians the size of angle AOB.

1

$$\angle AOB = \frac{\pi}{3} \checkmark$$

* Asked for RADIANS not degrees.

(b) Calculate the area of the shaded region. (Write your answer as an exact value)

3

$$= \text{area of sector} - \text{area of triangle}$$

$$= \frac{1}{2} r^2 \theta - \frac{1}{2} ab \sin C$$

* Learn use formula correctly
✓ 'Radians' vs Degrees

$$= \frac{1}{2} \times 15^2 \times \frac{\pi}{3} - \frac{1}{2} \times 15^2 \times \sin \frac{\pi}{3}$$

$$= \frac{225\pi}{6} - \frac{1}{2} \times 15^2 \times \frac{\sqrt{3}}{2} \checkmark$$

* Well done by most.

$$= \frac{75\pi}{2} - \frac{225\sqrt{3}}{4}$$

$$\text{OR } \frac{150\pi - 225\sqrt{3}}{4}$$

$$\text{OR } \frac{225}{2} \left[\frac{\pi}{3} - \frac{\sqrt{3}}{2} \right]$$

✓
Anything that was 'exact'

* Some didn't answer exact (-1)mk

Question 23

A uniform continuous random variable with probability density function is shown below.

$$f(x) = \begin{cases} ax^2, & 0 \leq x \leq 6 \\ 0, & \text{otherwise} \end{cases}$$

(a) Find the value of a .

2

$$\int_0^6 ax^2 dx = 1$$

$$\checkmark \left[\frac{ax^3}{3} \right]_0^6 = 1$$

$$\frac{216a}{3} - 0 = 1$$

$$216a = 3$$

$$a = \frac{3}{216}$$

$$\boxed{a = \frac{1}{72}}$$

* well done by most

(b) Find $P(3 \leq X \leq 5)$

3

$$\int_3^5 \frac{x^2}{72} dx$$

$$= \frac{1}{72} \left[\frac{x^3}{3} \right]_3^5$$

$$= \frac{1}{72} \times \left[\frac{125}{3} - \frac{27}{3} \right] = \frac{49}{108}$$

* some 'carry on' marks awarded

fraction + decimal + % accepted

(0.4537... also accepted)

Question 24

The circle $x^2 + 6x + y^2 - 4y - 3 = 0$ is reflected in the x -axis.

3

Sketch on the next page the reflected circle, showing the coordinates of the centre and the radius.

$$x^2 + 6x + y^2 - 4y = 3$$

$$(x+3)^2 + (y-2)^2 = 3 + 9 + 4$$

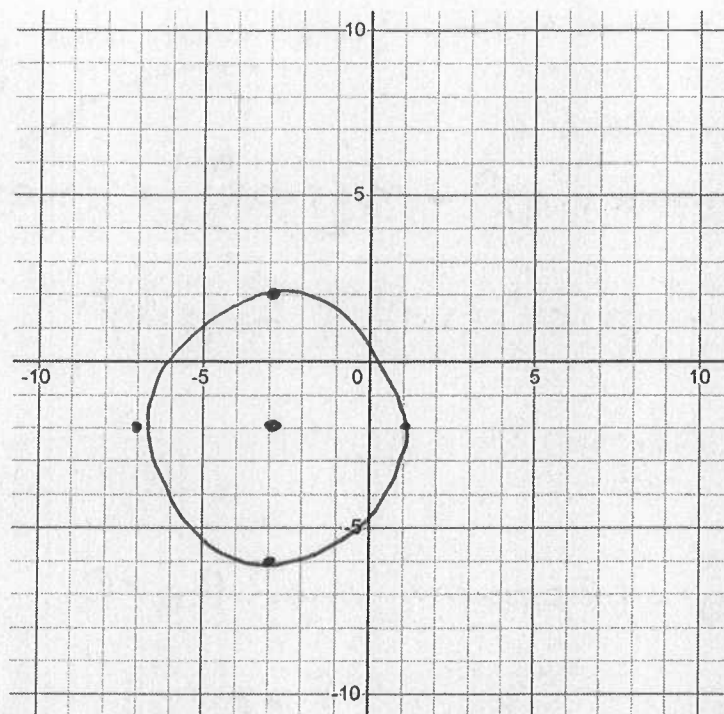
$$(x+3)^2 + (y-2)^2 = 16$$

centre $(-3, 2)$

radius $= 4$

Reflection in x axis centre $(-3, -2)$

✓
reflection
sketch



* had
to
cross
relevant
axis

* some
v. poor
sketches

Question 25

* many ^{incorrectly} reflected in y axis!

Jason deposits an amount of \$100 000 into an account which pays compound interest of 2% per annum, added to the account at the end of each year.

Immediately after the interest is added, Jason makes a withdrawal for expenses for the coming year. The first withdrawal is \$M. Each subsequent withdrawal is 10% greater than the previous one.

Let A_n be the amount of money in the account after the n th withdrawal.

(a) Show that $A_2 = 100\,000(1.02)^2 - M(1.02 + 1.10)$

1

$$\begin{aligned} A_1 &= 100\,000 \times 1.02 - m \quad \checkmark \text{ had to have this.} \\ A_2 &= [100\,000 \times 1.02 - m] \times 1.02 - m \times 1.10 \quad \left. \begin{array}{l} \text{\$} \\ \text{develop} \end{array} \right\} \\ &= 100\,000 \times 1.02^2 - m \times 1.02 - m \times 1.10 \\ &= 100\,000 \times 1.02^2 - m[1.02 + 1.1] \quad (\text{QED}) \end{aligned}$$

(b) Show that $A_3 = 100\,000(1.02)^3 - M[(1.02)^2 + (1.02)(1.10) + (1.10)^2]$

1

$$A_2 = 100\,000 \times 1.02 - m(1.02 + 1.1) \quad \text{from part a)}$$

$$\begin{aligned} A_3 &= [100\,000 \times 1.02 - m(1.02 + 1.1)] \times 1.02 - m \times 1.1^2 \\ &\quad \text{something like highlighted} \checkmark \\ &= 100\,000 \times 1.02^2 - m[1.02^2 + 1.1(1.02)] - m \times 1.1^2 \\ &= 100\,000 \times 1.02^2 - m[1.02^2 + 1.1(1.02) + 1.1^2] \end{aligned}$$

* some wrote out
part (a) solution
then expanded the A_3 given.

* Students need to
DEVELOP
the series not
fudge it!

* Difficult!! * this term here was the error for many.

(c) Hence, write an expression for A_n

1

$$A_n = 100\,000 \times 1.02^n - m \left[1.02^{n-1} + (1.02)^{n-2} (1.1) + \dots \dots (1.02)(1.1)^{n-2} + (1.1)^{n-1} \right]$$

(d) Show there will be NO money in the account when $M = \frac{8000}{\left[\left(\frac{1.10}{1.02} \right)^n - 1 \right]}$

2

No money in account when $A_n = 0$.

$$0 = 100\,000 \times 1.02^n - m \left[\begin{array}{l} \text{AP } a = 1.02^{n-1} \\ r = \frac{1.10}{1.02} \quad n = n \end{array} \right]$$

$$0 = 100\,000 \times 1.02^n - m \left[\frac{1.02^{n-1} \left[\left(\frac{1.10}{1.02} \right)^n - 1 \right]}{\frac{1.10}{1.02} - 1} \right]$$

$$m \left[\frac{1.02^{n-1} \left[\frac{1.10^n}{1.02^n} - 1 \right]}{\frac{1.10 - 1.02}{1.02}} \right] = 100\,000 \times 1.02^n$$

$$m \left[\frac{1.02^{n-1} \left[\frac{1.10^n}{1.02^n} - 1 \right]}{\frac{0.08}{1.02}} \right] = 100\,000 \times 1.02^n$$

$$m \left[\frac{1.02^{n-1} \left[\frac{1.10^n}{1.02} - 1 \right]}{0.08} \right] \times \frac{1.02}{1.02} = 100\,000 \times 1.02^n$$

$$M \left[1.02^n \left[\frac{1.10^n}{1.02^n} - 1 \right] \right] = 8000 \times 1.02^n$$

$$M = \frac{8000 \times \cancel{1.02^n}}{\cancel{1.02^n} \left[\left[\frac{1.10}{1.02} \right]^n - 1 \right]}$$

$$M = \left\{ \frac{8000}{\left[\frac{1.10}{1.02} \right]^n - 1} \right\}$$

* Very tricky Question

* 1 mark awarded for adequate progress

* No marks for 'fudging'

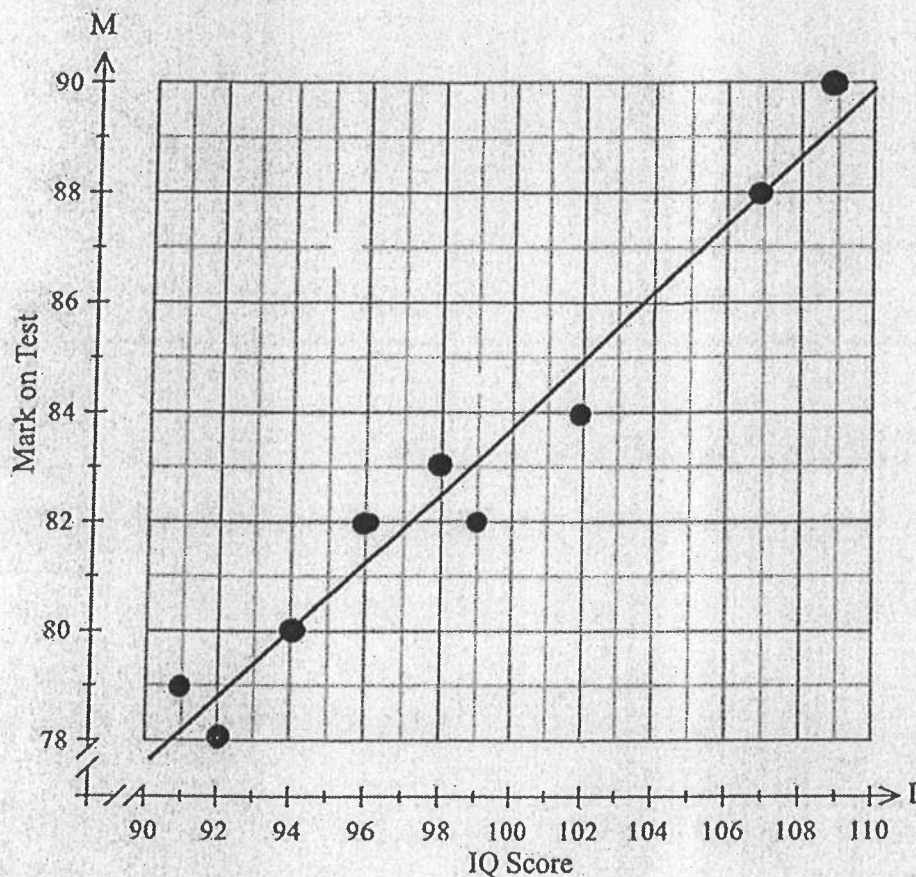
* needed to show a series

sum where $r = \frac{1.10}{1.02}$

for 1 mark here

Question 26

The bivariate data in the scatterplot below compare the recorded IQ scores of students with their mark on a class test out of 100. A line of best fit was also drawn.



(Note: Both axes are truncated)

(a) Calculate the gradient of the line of best fit

1

$$m = \frac{88 - 80}{107 - 94} = \frac{8}{13} = 0.61538 \dots$$

Accepted $\frac{5}{8}$ or any answer close to 0.6

(b) Describe the correlation between the IQ score and the mark on the test in terms of **direction** and **strength**.

2

Strong Positive.

Question 27

A continuous probability distribution is given by $f(x) = \frac{3x^2}{279}$ defined in the domain $[4, 7]$

(a) Find the cumulative distribution function.

3

$$\begin{aligned} \int_4^x \frac{3x^2}{279} dx &= \frac{3}{279} \int_4^x x^2 dx \\ &= \frac{3}{279} \times \left[\frac{x^3}{3} \right]_4^x = \left[\frac{x^3}{279} \right]_4^x \\ \therefore &= \frac{1}{279} (x^3 - 4^3) = \frac{x^3 - 64}{279} \end{aligned}$$

(b) Find the median of the continuous probability distribution.

2

For median $CPD = 0.5$

$$\therefore \frac{x^3 - 64}{279} = 0.5$$

$$x^3 = 0.5 \times 279 + 64 = 203.5$$

$$x = 5.88...$$

Question 28

Evaluate $\int_1^3 x^2(x^3+8)^2 dx$

By formula $\int f(x)[f(x)]^n dx = \frac{[f(x)]^{n+1}}{n+1} + C$

$$= \frac{1}{3} \int_1^3 3x^2(x^3+8)^2 dx$$

$$= \left[\frac{1}{3} \times \frac{(x^3+8)^3}{3} \right]_1^3 = \frac{1}{9} (42146)$$

$$\frac{d}{dx} (x^3+8)^3 = 3(x^3+8)^2 \cdot 3x^2 = 9x^2(x^3+8)^2$$

$$\therefore \frac{1}{9} \int \frac{d}{dx} (x^3+8)^3 dx = \left[\frac{1}{9} (x^3+8)^3 \right]_1^3$$

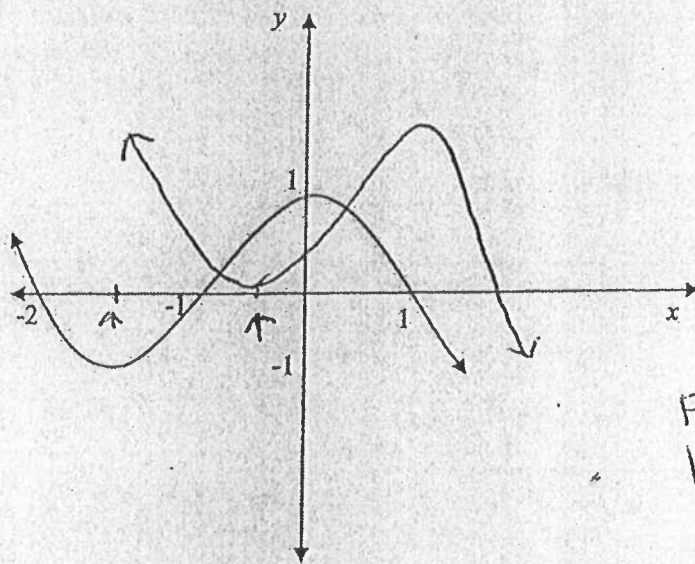
$$= \frac{1}{9} (42875 - 729)$$

$$= \frac{42146}{9}$$

In decimals (4682.89)
2dp.

Question 29

(a) The graph of $y = f(x)$ is shown.



Shift Right 1
translate up 1

For full marks 1st t.p.
had to be above/on
x-axis.

On the graph above draw the curve, $y = f(x-1)+1$, showing all important features.

2

Question 30

A class of students sat a test worth 30 marks. The marks were normally distributed.

The mean was 18 and the standard deviation was 1.6.

(a) Calculate Julie's z-score if her mark was 22.

1

$$z = \frac{x - \mu}{\sigma} = \frac{22 - 18}{1.6} = 2.5$$

(b) If Dan's z-score was 1.25, calculate his actual mark.

1

$$1.25 = \frac{x - 18}{1.6}$$

$$2 = x - 18$$

$$x = 20$$

Question 31

A psychologist proposes that the ability of a child to memorise during the first five years can be modelled by the function $f(x) = 1 + x \log_e x$ $0 < x \leq 5$

(i.e the ability to memorise at age x years is $f(x)$)

(a) During which month is the ability at a minimum in the first five years

3

$$\begin{aligned}
 f(x) &= 1 + x \ln x & \frac{d}{dx}(x \ln x) &= \ln x + 1 \\
 f'(x) &= \ln x + 1 & u &= x \quad v = \ln x \\
 \text{for s.p. } f'(x) &= 0 & u' &= 1 \quad v' = \frac{1}{x} \\
 \ln x + 1 &= 0 & \therefore \frac{d}{dx}(x \ln x) &= \ln x + 1 \\
 \log_e x &= -1 & & \\
 e^{-1} &= x = 0.367 \dots \text{ yrs.} & & \\
 \text{for month value } x &= 12 & & \\
 &= 4.41 \text{ months} & & \\
 \therefore & \text{5th month.} & &
 \end{aligned}$$

(b) When is it a maximum during this period. Give a reason for your answer.

2

$$\begin{aligned}
 & \text{Maximum occurs when } x = 5 \\
 \text{Graph:} & \quad \left| \begin{aligned} & f(x) \text{ is a continuous fn.} \\ & f'(x) > 0 \text{ (inc)} \\ & \text{Graph: } x = 5 \\ & 1 \text{ min t.p. } f''(x) = \frac{1}{x} \\ & f'\left(\frac{1}{e}\right) = 2.71 > 0 \therefore \text{MIN} \\ & \text{Possible reasons:} \end{aligned} \right.
 \end{aligned}$$

Question 32

Evaluate $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y$, given that $y = xe^{-2x}$.

$$y = xe^{-2x}$$

$$y' = -2xe^{-2x} + e^{-2x}$$

$$y'' = -2x \cdot -2e^{-2x} + e^{-2x} \cdot -2 - 2e^{-2x}$$

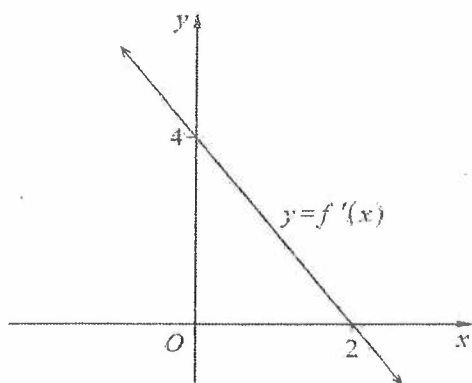
2

$$y'' + 4y' + 4y = 4xe^{-2x} + 4e^{-2x} - 8xe^{-2x} + 4e^{-2x} + 4xe^{-2x} = 0$$

Question 33

The graph of the derivative of a function $y = f'(x)$ is shown below.

3



$$f'(x) = mx + b$$

$$b = 4 \quad m = -2$$

$$\therefore f'(x) = -2x + 4$$

$$\therefore f(x) = -x^2 + 4x + C$$

The curve $y = f(x)$ has a maximum value of 12.

What is the equation of $y = f(x)$?

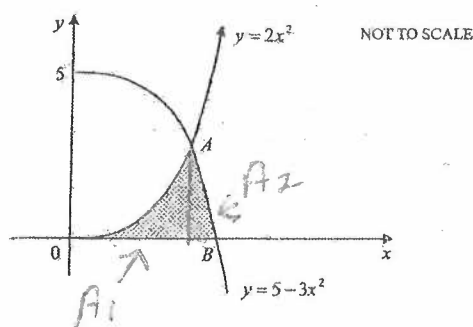
$$\text{Max occurs when } x = 2 \text{ and } y = 12$$

$$12 = -4 + 8 + C$$

$$\therefore C = 8$$

$$\therefore y = -x^2 + 4x + 8$$

Question 34



The shaded region OAB is bounded by the parabolas $y = 2x^2$, $y = 5 - 3x^2$ and the x-axis.

Point A is the intersection of the two parabolas and point B is the x-intercept of the parabola $y = 5 - 3x^2$

- (a) Show that the x-coordinates of A and B are 1 and $\frac{\sqrt{5}}{\sqrt{3}}$ respectively.

2

① $2x = 5 - 3x^2$

$5x^2 = 5$

$x = \pm 1$

$\therefore A$ is $x = 1$

② $5 - 3x^2 = 0$

$x^2 = \frac{5}{3}$

$\therefore B$ is $x = \frac{\sqrt{5}}{\sqrt{3}}$

1mk for EACH POINT CLEARLY SHOWN

- (b) Find the exact area of the shaded region OAB.

3

A① $\int_0^1 2x^2 dx$

$= \left[\frac{2x^3}{3} \right]_0^1$

$= \frac{2}{3} u^2$

A② $\int_1^{\frac{\sqrt{5}}{\sqrt{3}}} (5 - 3x^2) dx$

$= \left[5x - x^3 \right]_1^{\frac{\sqrt{5}}{\sqrt{3}}}$

$= \left(\frac{5\sqrt{5}}{\sqrt{3}} - \frac{5\sqrt{5}}{3\sqrt{3}} \right) - (4)$

$= \frac{10}{3} \frac{\sqrt{5}}{\sqrt{3}} - 4$

1mk EACH

$\therefore \text{TOTAL} = A_1 + A_2$

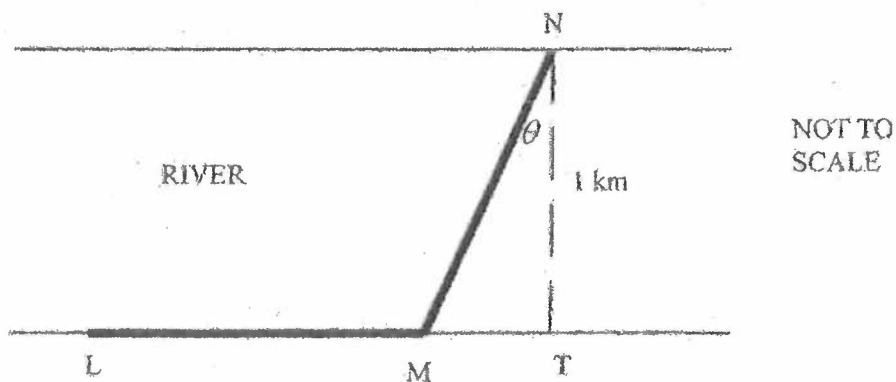
$= \frac{10}{9} [\sqrt{15} - 3] u^2$ OR EQUIVALENT

1mk for correct answer.

Question 35

It is desired to construct a cable link between two points L and N, which are situated on opposite banks of a river of width 1 km. L lies 3 km upstream from N. It costs 3 times as much to lay a length of cable underwater as it does to lay the same cable overland.

The following diagram is a sketch of cables where θ is the angle where NM makes with the direct route across the river.



Note: $MN = \sec \theta$ and $MT = \tan \theta$

(a) If segment LM costs c dollars per km, prove the total cost (T) of laying the cable

2

is given by $T = 3c - c \tan \theta + 3c \sec \theta$

$$LM = LT - MT$$

$$LT = 3 \quad MT = \tan \theta$$

$$LM = 3 - \tan \theta$$

$$\text{Cost of LM} = 3c - c \tan \theta$$

link for EACH
segment (LM & MN)

Clearly

EXPLAINED

$$MN = \sec \theta$$

$$\text{Cost of MN} = 3c \sec \theta$$

!!!

$$\therefore T = 3c - c \tan \theta + 3c \sec \theta$$

(b) At what angle should the cable cross the river in order to minimise the total cost of laying it. 3

$$T = 3c - c \tan \theta + 3c \sec \theta$$
$$= 3c - c \tan \theta + 3c (\cos \theta)^{-1}$$

$$T' = -c \sec^2 \theta - 3c (\cos \theta)^{-2} \cdot \sin \theta$$
$$= -c \sec^2 \theta + 3c \frac{\sin \theta}{\cos^3 \theta} \sec \theta$$
$$= -c \sec^2 \theta + 3c \tan \theta \sec \theta$$
$$= -c \sec^2 \theta + 3c \frac{\sin \theta}{\cos^2 \theta}$$

} link for one of these

Let $T' = 0$

$$0 = -\frac{c}{\cos^2 \theta} + 3c \frac{\sin \theta}{\cos^2 \theta}$$

$$0 = -c + 3c \sin \theta$$

$$\sin \theta = \frac{1}{3}$$

$$\theta = 19^\circ 28' (0.34^\circ)$$

link for correct angle

Test: $\theta = 19^\circ$ $T' < 0$ $\checkmark$

$$\theta = 20^\circ \quad T' > 0$$

link for correct TEST

$$\therefore \theta = 19^\circ 28' \text{ gives minimum}$$

End of Exam